

THE STRUCTURE OF MATHEMATICAL REALITY

Phase III — Chapter 10

Dynamics and Approximation: The Structure of Change

Lecture

Self-Study Curriculum

Every model you have ever built that runs forward in time is a dynamical system. A population growing, a drug clearing from plasma, a signal relaxing to baseline, a gene switching on: each is a rule for how a state changes, iterated forward. This is the field your mechanistic instinct has been living in without a name. The flagship example of this whole curriculum, that the *same* differential equation governs radioactive decay, population decline, and capacitor discharge, is not really a fact about algebra or about analysis in the abstract. It is a fact about *dynamics*, about the structure of change itself. Chapter 1 put “analysis studies limits, what happens as you approach, approximate, and pass to infinity” on your map and then, deliberately, never developed it. This chapter develops it, because dynamics is where the mathematics of change and the mathematics of *approximation* both live, and approximation is the missing half of this curriculum’s central answer to why mathematics works at all.

Topology, in Chapter 9, handed you a static stage: a state space with a certain shape, its holes and connectedness and compactness frozen in place. This chapter sets the stage in motion.

State, and the space of all states

The central move of dynamics, and the one that reorganizes everything, is to stop thinking about individual *trajectories* (this one population over these ten years) and start thinking about the *rule* that produces every trajectory at once.

A dynamical system has exactly two ingredients.

1. A **state**: the minimal set of numbers you need to know about the system *right now* to determine its entire future. This word *minimal* is doing real work. For a decaying isotope the state is one number, the amount remaining, because that alone fixes what happens next. For a pendulum the state is two numbers, the angle *and* the angular velocity: position alone does not determine the future, because a pendulum passing through the bottom moving left and one passing through the bottom moving right do completely different things next. The state is whatever you must carry so that the future depends only on the present, not on the history of how you arrived. Engineers call this the Markov property; it is the mathematical statement that the present is a sufficient summary of the past.
2. A **rule of evolution**: a law for how the state changes. In continuous time this is a differential equation, $dx/dt = f(x)$. The function f assigns to every possible state a *velocity*: the direction and speed the system moves next, given where it is now.

The set of all possible states is the **state space** (or phase space). Now the reframing that changes everything. *The rule f is a vector field on the state space.* At every point it plants an arrow: “if you are here, go this way, this fast.” A trajectory is simply what you trace by dropping a marble somewhere and letting it follow the arrows. You are no longer studying one history among many. You are studying the **flow**: the entire field of all possible histories at once. This is the “canvas” of Chapter 8 set in motion. The state space is the canvas; the vector field is the structure painted on it; a trajectory is one brushstroke among infinitely many the field already determines.

A worked example to anchor this. Consider a two-compartment model of a drug: amount in blood and amount in tissue. The state is the pair (blood, tissue), two numbers, and the state space is the quarter-plane of non-negative amounts. The rule is a pair of coupled equations: the drug flows from blood to tissue at a rate proportional to how much more is in blood, back at a rate proportional to the tissue level, and out of blood by clearance. That rule is a vector field: stand at any point (blood, tissue) and it tells you the instantaneous change in both. One

patient's time course is one trajectory. But the field describes *every* patient at *every* starting dose simultaneously, and the shape of the field, not any single curve, is what you actually want to understand. *What this adds to your map:* you have traded a stack of individual time series for a single geometric object, the flow, and from here on every question about the system is a question about the geometry of that flow.

Fixed points and stability: where the structure lives:

Most of the qualitative behavior of a system is decided by a small number of special states, and finding them is the first thing to do with any flow.

A **fixed point** (equilibrium) is a state where $f(x) = 0$: the velocity vanishes, so a system placed exactly there stays exactly there forever. Equilibria are the skeleton of the state space, the places where nothing moves. But not all equilibria are alike, and the distinction is the one your work turns on constantly.

- A **stable** fixed point pulls nearby states toward it. Perturb the system a little and it returns. Picture a marble at the bottom of a bowl.
- An **unstable** fixed point pushes nearby states away. The smallest perturbation grows without bound. Picture a marble balanced on the top of a dome.
- A **saddle** is stable along some directions and unstable along others: the marble on a mountain pass, which rolls back if pushed along the ridge but rolls away if pushed toward either valley.

Here is the point that beginners miss and that decides everything downstream: *stability is not something you read off the equilibrium value*. Two systems can have an equilibrium at exactly the same place, one stable and one unstable. Stability is a property of the flow *around* the point, of what the neighboring arrows do, whether they converge on the point or flee it. And “what the neighboring arrows do” is precisely a statement about the system near equilibrium, which is where approximation walks in.

A worked example you will do again in the exercises. Take the logistic equation for a population with growth rate r and carrying capacity K :

$$dN/dt = rN(1 - N/K).$$

Set the right side to zero. Either $N = 0$ (extinction) or $N = K$ (the population sits exactly at carrying capacity). Those are the two fixed points, the skeleton of this one-dimensional flow. Now sketch the arrows on the N -axis: for small positive N the growth rate is positive, so arrows point right, away from 0 and toward K ; for N above K the rate is negative, so arrows point left, back toward K . Extinction is unstable (any seed population grows away from it), and carrying capacity is stable (the population is pulled back to K from either side). You did not need to solve the equation to learn this. You read the stability straight off the direction of the arrows near each fixed point. That reading is what we now make systematic.

Linearization: why linear methods are unreasonably effective:

Chapter 5 promised that linearity is “approximately valid for small perturbations around any equilibrium,” and gave the claim a single paragraph. Here is the machinery behind that paragraph, because it is the missing half of this curriculum's central answer to why mathematics is so effective.

Near a fixed point x^* , write the state as $x^* + \delta$, where δ is a small displacement from equilibrium. Taylor's theorem, the workhorse of all local approximation, says that any smooth function f can be expanded around x^* as

$$f(x^* + \delta) = f(x^*) + f'(x^*) \cdot \delta + (\text{terms of order } \delta^2 \text{ and smaller}).$$

At a fixed point $f(x^*) = 0$, so the leading behavior of the perturbation is

$$d\delta/dt \approx f'(x^*) \cdot \delta.$$

Look at what happened. The evolution of the small displacement is governed by a *linear* equation whose coefficient is the constant number $f'(x^*)$. In more than one dimension that coefficient becomes the **Jacobian matrix**, the matrix of all partial derivatives of the components of f , evaluated at the fixed point; but the story is the same. The perturbation grows or decays exponentially, and the rates are set by the **eigenvalues** of that matrix. Negative real part means the displacement shrinks, so the fixed point is stable; positive real part means it grows, so the point is unstable; a nonzero imaginary part means the displacement rotates as it grows or shrinks, which is oscillation. Run this on the logistic equation: $f'(N) = r - 2rN/K$, so $f'(0) = r > 0$ (extinction unstable, perturbations grow at rate r) and $f'(K) = -r < 0$ (carrying capacity stable, perturbations decay at rate r). The signs you read off the arrows a moment ago are exactly the signs of these derivatives. The picture and the calculation agree because they are the same fact.

Now stop and see what this means, because it is one of the deepest ideas on your map. Chapter 3 flagged the eigenvalue as living at the intersection of algebra, geometry, and analysis, without saying why. Here it appears in its *analytic* role: the eigenvalue is the *rate constant* of the flow near equilibrium, the number that sets how fast perturbations relax or blow up. And the reason your linear tools, eigendecomposition, linear stability analysis, small-signal approximation, work at all is now stateable in one sentence. It is not that the world is linear. It is that the world is *smooth*, and Taylor's theorem says a smooth system, viewed closely enough around an equilibrium, *is* linear. This is arguably the single deepest reason linear algebra is so disproportionately powerful relative to how little it assumes: linearity is the first-order shadow that every smooth system casts near its fixed points, and first order is usually enough.

This is also the precise, formal version of an intuition you brought to this curriculum yourself. You suspected that certain operations are powerful because they run along the natural grain of a problem, and that linearity is not a property of a system but a property of the *relationship between a system and its canvas* (Chapter 8). Linearization is that suspicion made exact. Near an equilibrium, the displacement coordinate δ *is* the right canvas, handed to you automatically: in the δ coordinate the leading dynamics obey superposition no matter how baroque f was in the original variables, because Taylor's theorem keeps only f 's linear part there. Keep this operator sense of "linear," a rule that respects adding and scaling states, separate from the everyday sense in which plotting $\log N$ straightens an exponential growth curve. The log canvas changes only how the trajectory draws; the δ canvas changes whether the rule superposes, and it is the second that stability turns on. The engineer's three questions, how large an error can I tolerate, how much stability margin do I have, how big a perturbation before the approximation breaks, are then the same question seen three ways, because error, stability margin, and perturbation size are all just the magnitude of δ . That family of questions finally has a home on your map.

One more consequence, because it explains something you have surely noticed and it points to the next chapter. Two systems with completely different mechanisms often obey the *same* linear equation near equilibrium, and after linearization this is not surprising but expected: linearization discards everything except the Jacobian, so any two smooth systems that happen

to share a Jacobian are, to leading order, indistinguishable. This is exactly why population decline near extinction reduces, to leading order, to $d\delta/dt = (\text{constant}) \cdot \delta$, the flagship form from Chapter 1. Radioactive decay and capacitor discharge reach the same equation by a shorter road: they *are* that equation exactly and globally, because in each the rate really is proportional to the amount, so no linearization is needed. Exactly or only to leading order, the mechanisms are unrelated and the *local structure of the flow* is identical. Chapter 11 will take this observation and blow it up into a general phenomenon, universality, in which whole classes of microscopically different systems flow to the same effective description; and it will turn out that the fixed point you are studying here reappears there, living in the space of *theories* rather than the space of states.

When linearization fails: the genuinely nonlinear phenomena:

Linearization is a local tool, and honesty about its scope is the whole discipline. It tells you everything *near* an equilibrium and, by construction, nothing about behavior far from one, because the terms of order δ^2 and smaller that it threw away are exactly what dominate once δ is no longer small. The phenomena that define biology live precisely where linearization runs out. There are four of them worth naming, and each is genuinely nonlinear in the sense that no linear model can produce it.

Multistability. Two or more stable fixed points coexist, and which one the system settles into depends on where it started. The state space divides into *basins of attraction*, one valley draining to each stable point, and the ridges between the basins decide the fate of any initial condition. This is the structure behind cell-fate decisions, genetic toggle switches, and hysteresis: a system that “remembers” because its landscape has more than one valley. A committed cell does not average its options; it sits in one basin, and returning it to the other requires a push over the ridge, not a nudge. A linear system cannot do this at all: a linear system has exactly one equilibrium (generically), so it can offer only one destiny. Multistability requires the nonlinear terms that carve a second valley. A concrete carrier is a genetic toggle: two genes whose products each repress the other. Whichever gene gets ahead suppresses its rival and locks in, so the system has two stable states (gene A high, or gene B high) separated by an unstable threshold state between them. Start slightly on one side and you commit to one fate; start on the other and you commit to the opposite; and once committed, ordinary fluctuations cannot flip you, because they are too small to climb the ridge between the basins. That is cellular memory implemented as landscape, and it is unreachable for any linear model, which has no second valley to offer.

Oscillation. A **limit cycle** is a closed loop in state space that nearby trajectories spiral onto: a stable *rhythm* rather than a stable *point*. Circadian clocks, heartbeats, predator-prey cycles, and gene-expression oscillators are limit cycles. The word *stable* matters: perturb the rhythm and it returns to the same loop, the same amplitude and period, just as a perturbed marble returns to the bottom of its bowl. This is genuinely beyond linear reach. A purely linear system can oscillate only at a knife’s edge, at the exact parameter value where a pair of eigenvalues has zero real part, and such oscillation is fragile: its amplitude is set entirely by the initial kick and drifts with the slightest perturbation. An *attracting* rhythm with its own preferred amplitude is a nonlinear object. The way one is typically born connects straight back to linearization and to Exercise 3: as you turn a parameter, a complex-conjugate pair of eigenvalues of the Jacobian crosses from negative real part to positive, the equilibrium loses stability, and a small limit cycle blooms around it. That event is a **Hopf bifurcation**, and if you linearize a real system and find a complex pair with small positive real part, you should suspect you are just past one and expect a growing oscillation as you push the parameter further.

Bifurcation. A *qualitative* change in the flow as a parameter varies. Turn a knob slowly and for a long while nothing structural happens, the fixed points sit where they were, stability unchanged; then at a critical value a fixed point appears, or vanishes, or swaps stability, and the system’s entire behavioral repertoire reorganizes at once. Thresholds, tipping points, and the onset of oscillation are all bifurcations. This is the structural meaning of the word “threshold” that a linear model can never carry: a linear response is proportional everywhere, so it has no special value at which behavior reorganizes, whereas a bifurcation *is* that special value. This is also exactly the “structural misspecification” failure that Chapter 14 will name for models, now seen from the inside: a model with no bifurcation fitted to a system that has one is not merely inaccurate, it is the wrong *kind* of object, and it will be confidently wrong precisely at the tipping point you most need to predict. The cleanest biological instance is an epidemic threshold. In the simplest transmission model the infected fraction has an equilibrium at zero (no outbreak), and its stability is governed by the basic reproduction number, the average number of new infections one case produces. Linearize around the disease-free state and the growth rate of a small outbreak is proportional to that number minus one. Below one, the zero-infection equilibrium is stable and introductions die out; above one, it turns unstable and a new endemic equilibrium appears. The crossing at exactly one is a bifurcation, a transcritical one, where two equilibria meet and exchange stability, and it is the entire meaning of a herd-immunity threshold. A purely statistical curve fitted to case counts on one side of that value carries no knowledge that the other side even exists; the dynamical view predicts the reorganization, and predicts where it sits, from the sign of a single linearized rate.

Chaos. Deterministic yet unpredictable. A chaotic system obeys an exact rule with no randomness in it, and yet nearby trajectories separate exponentially fast, a property called *sensitive dependence on initial conditions*: the tiniest uncertainty in the present is amplified until, after enough time, you know nothing about the state. The rule is short and the long-range behavior is not compressible into any long-range prediction. Chaos is the sharpest possible demonstration that “we know the equations” and “we can predict the future” are *different claims*, a distinction that will matter a great deal when you reach Chapter 15 and the question of what it costs to describe a system. You can have the generating rule in full and still be unable to say where the system will be next month, not for lack of computing power but in principle, because any error you start with, however small, is magnified without bound. The textbook carrier is the discrete logistic map, the same growth rule iterated in discrete steps: raise its growth parameter and the long-run behavior passes from a single stable level, to a two-step oscillation, to four, and then into a regime where the sequence never repeats even though the rule generating it is a single line of arithmetic. Population counts, cardiac rhythms, and neural firing all have regimes of this kind, which is why a clean deterministic mechanism can produce a time series indistinguishable at a glance from noise. Telling the two apart is then not a matter of staring harder at the trace but of asking a structural question: does the signal live on a low-dimensional attractor, or does it fill its space the way genuine randomness would?

A structural takeaway that ties three of the exercises together. Multistability, oscillation, and bifurcation are the vocabulary for the failures of the linear worldview, and each answers a question the exercises pose. A “threshold” system that does nothing below some input and switches on sharply above it (Exercise 2) cannot be a linear model, structurally, because linearity is proportionality with no special value; the phenomenon that *can* represent it is a bifurcation, and to locate the threshold you must find the critical parameter value at which the flow reorganizes. A complex eigenvalue pair with small positive real part (Exercise 3) says the system is just past a Hopf bifurcation and should be developing an attracting oscillation. And distinguishing a

chaotic signal from genuine noise (Exercise 4) turns on determinism: a low-dimensional chaotic system, replotted in state space rather than against time, reveals structure, a folded-up attractor of finite dimension, and is *short-term* predictable even though it is long-term unpredictable, whereas true noise has neither the low-dimensional geometry nor the short-term forecastability. Determinism does not buy you predictability, but it does buy you structure you can sometimes detect.

Feedback: the mechanism behind all of it:

Everything above is the mathematics of **feedback**, and this is the part your engineering instinct already trusts completely; it needs a translation, not an introduction. A term in $f(x)$ that depends on x is a feedback loop: the state feeds back into its own rate of change. The sign of that dependence is the whole story, and it is the same information as the sign of the Jacobian.

- **Negative feedback**, where the rate opposes the displacement, builds stable fixed points. This is homeostasis: push the system away and it drives itself back. A gene that represses its own production settles quickly to a set level and resists perturbation.
- **Positive feedback**, where the rate reinforces the displacement, builds instability, switches, and multistability. A gene that activates its own production, once past a threshold, runs away to a high state and stays there; two such genes repressing each other give the toggle switch and its two stable cell fates.
- **Fast negative feedback combined with slow positive feedback** builds oscillation: the positive loop pushes the system up, the delayed negative loop pulls it back, and the mismatch in timing turns what would be a resting point into a rhythm. This is the generic architecture of biological clocks.

You do not need new intuition for any of this. You need only to see that “feedback loop” and “the sign of the Jacobian” are the same fact stated in two languages, the mechanist’s and the mathematician’s, and that translating freely between them is most of what dynamical analysis is.

One more structural consequence, which answers a question the quiz will press. Suppose you know a system’s phase space is bounded (trajectories cannot run off to infinity) and that its only equilibrium is unstable (trajectories are pushed away from the one resting place). What must happen in the long run? The trajectory cannot leave the bounded region, and it cannot come to rest at the one fixed point it is repelled from, so it has nowhere to settle and yet nowhere to escape. It is *trapped* and *restless*. In the plane, the only remaining option is that it winds onto a closed loop: a limit cycle must exist. This trapped-trajectory argument (formally the Poincaré-Bendixson conclusion for two-dimensional flows) lets you *predict a rhythm from structure alone*, without solving anything, just as compactness in Chapter 9 let you predict that an optimum exists without computing it. Boundedness plus a single unstable equilibrium forces sustained oscillation.

Where you now stand:

Analysis, back in Chapter 1, “studies limits, what happens as you approach, approximate, and pass to infinity.” That was an abstract charge with no content attached. Dynamics is where it becomes concrete and mechanistic, and this chapter has now spent all three of its limits. The long-time limit of a trajectory is the question of whether the system settles, cycles, or wanders (fixed points, limit cycles, chaos). The small-perturbation limit is linearization, the first-order shadow that makes linear algebra unreasonably effective near equilibrium. The critical-parameter limit

is bifurcation, the reorganization of the whole flow at a threshold. Approximation, named as the missing half of this curriculum's answer, turned out to be Taylor's theorem doing its quiet work: the world is modelable near its equilibria because smoothness makes it locally linear, and it stops being modelable by linear means exactly where the discarded higher-order terms take over.

What this adds to your map is the grammar of change itself. A dynamical system is a flow on a state space; its behavior is organized by fixed points, stability, and bifurcations, and, when the flow is nonlinear, by multistability, oscillation, and chaos. When you next hit a wall in your real work, it will most often be a nonlinear-dynamics wall: a threshold you cannot linearize across, a switch, an oscillation, a regime that will not sit still. You now have the words for those walls, and the words tell you which tool applies and which cannot.

The question this chapter hands forward is about the limits of the deterministic picture. You have seen that a system can be fully determined and still unpredictable (chaos), and you have seen the same fixed point govern mechanisms that share nothing but their local structure. Both hints point past determinism toward the mathematics of many systems at once and of what survives when you average over detail. Chapter 11 takes that step: it asks what probability *is*, why averaging works at all, and why microscopically different systems, coarse-grained, flow to the *same* effective description, the same fixed point now living in the space of theories rather than the space of states. The flow you learned to read here is about to reappear one level up, deciding not where a single system goes, but which laws whole classes of systems are funneled toward.